Sales Performance Analysis Report: 2020 – 2021

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Summary

This report analyzes sales performance for Velora Stores from January 2020 to July 2021 using Power BI. The objective was to evaluate business performance, identify key revenue and demand drivers, and uncover optimization opportunities across products, customers, channels, and regions.

Results show that 2020 delivered strong but volatile performance, generating $7.08M in revenue, driven by a small group of high-performing products and regions. In contrast, 2021 experienced a significant decline, with revenue falling to $3.48M (−51% YoY), alongside reduced order volumes, quantity sold, and Average Order Value. The sharp −99% drop in July 2021 suggests operational or data-related disruption rather than a gradual loss of demand.

The analysis highlights heavy revenue concentration, declining customer value, and structural regional performance differences. Recommendations focus on validating operational continuity, strengthening core products, improving pricing and basket size, prioritizing high-margin customers and channels, and using predictive analysis in Power BI to support data-driven decision-making.

BUSINESS OBJECTIVES

This project analyses sales data to assess business performance, demand patterns, and uncover optimization opportunities across customers, products, and regions.

Specifically, this analysis aims to:

- Evaluate overall sales performance and demand trends across time to assess business health and growth.
- Identify the key drivers behind sales results and highlight areas where performance can be optimized.
- Analyze customer purchasing behavior to understand who is buying, where performance is strongest, and where weaknesses exist across regions, products, or channels.

- Deliver actionable insights that support data-driven decision-making and strategic planning.

DATA OVERVIEW

- Data Source: Store sales database (Excel)
- Time Period: Jan 2020 – July 2021
- Records Analyzed: 8,000 customer data entries
- Key Variables/KPIs: Sales Revenue, Total Completed Orders, Total Quantity Sold, and Average Order Value (AOV).

METHODOLOGY AND DATA PREPARATION

The analysis was conducted using Power BI.

Data preparation steps included:

- Created three additional dimension tables: Customers, Orders, and Date, to establish a star schema data model.
- Removed unnecessary columns.
- Merged queries in the Order_Fact table and Customer_Dimension table.
- Created new columns: Unique Customer ID, Month, Half Month, Month Number, Quarters, and Year.
- Removed duplicates.
- Organized the Month column in the Date table.
- Validated and adjusted data types.

KEY FINDINGS

OVERVIEW

Total sales and order volumes trends over time (monthly and yearly)

Year 2020

- Total sales revenue: $7.08M
- Total quantity sold: 170,346
- Total orders: 5,295

Monthly Trend (MoM Growth) in 2020

- January recorded revenue of $500.63K, quantity sold of 12,527, and orders of 391.
- In February, there was an 8.98% positive growth rate in revenue, a 7.56% increase in quantity sold, and a 5.37% increase in orders.
- March also recorded an increase, although there was a decline in the growth of sales revenue (3.06%). Quantity sold increased by 7.3%, while orders increased by 9.71%.
- In April, sales revenue increased significantly by 16.01%. Quantity sold dropped (-0.98%), while orders remained the same (0% growth).
- May witnessed negative revenue growth (-3.7%). Quantity sold adjusted positively by 2.13%, while orders declined (-1.55%).
- All three KPIs recorded negative growth in June.
- July showed slight recovery: sales revenue grew by 5.81%, quantity sold rose by 2.44%, while orders dropped (-0.45%).
- August improved: revenue increased by 7.93%, quantity sold peaked at 15.83%, and orders rose significantly by 12.53%.
- September declined: revenue (-6.4%), quantity sold (-15.63%), and orders (-13.56%).
- October increased: revenue (+3.97%), quantity sold (+4.85%), and orders (+7.73%).
- November: revenue increased (+5.6%), but quantity sold (-5.82%) and orders (-6.52%) declined.
- December: revenue recorded negative growth (-0.56%), while quantity sold (+8.19%) and orders (+5.12%) grew positively.

Year 2021

- Total sales revenue: $3.48M
- Total quantity sold: 85,995
- Total orders: 2,705

Monthly Trend (MoM Growth) in 2021

- January: revenue (-0.23%), quantity sold (+0.2%), orders (+3.76%).
- February: all KPIs declined — revenue (-10.88%), quantity sold (-8.71%), orders (-7.46%).
- March: positive growth — revenue (+1.76%), quantity sold (+10.55%), orders (+6.45%).
- April: negative growth — revenue (-5.39%), quantity sold (-7.54%), orders (-5.19%).

- May: significant positive growth — revenue (+16.63%), quantity sold (+9.68%), orders (+4.79%).
- June: negative growth — revenue (-14.39%), quantity sold (-8.81%), orders (-4.36%).
- July: -99% negative growth across all KPIs.

2. Products, revenue generated and order volume

2020 Top 5 Products:

- Chocolate Truffle: revenue $1.67M, quantity 9,868, orders 301
- Nougat: revenue $870.62K, quantity 10,266, orders 301
- Turkish Delight: revenue $543.79K, quantity 8,232, orders 250
- Brittle: revenue $534.46K, quantity 9,184, orders 288
- Toffee: revenue $540.02K, quantity 9,355, orders 280

2021 Top 5 Products:

- Chocolate Truffle, Nougat, Brittle, Toffee, Turkish Delight remained top contributors.
- In July (last transaction month), only four products sold: Nougat ($1.90K), Pudding ($663.97), Geplak ($538.61), Marshmallow ($254.27), each with one order.

3. Average order value (AOV), and variation amongst customers

- 2020 AOV: $928.90
  - Highest AOV range: $20–$28
  - Lowest AOV range: $13–$16

- 2021 AOV: $456.67
  - Highest AOV range: $10–$11
  - Lowest AOV range: $5–$7

PERFORMANCE

4. Sales performance across sales channels

- AXW291: $3.3M (2020), $1.63M (2021)
- GUT930: $1.6M (2020), $0.78M (2021)
- NXH382: $1.4M (2020), $0.71M (2021)
- FLR025: $0.7M (2020), $0.36M (2021)

5. Relationship between order quantities and sales values relate

Overall, there is a positive relationship between sales volume and revenue.

However, high-volume orders do not always translate to high revenue.

Example (2020):

Ei Group: 149 orders, 4,644 units, $188,678 revenue.

Prasco Group: 129 orders, 4,362 units, $219,104 revenue.

6. Products and sales trends

In 2021, all products experienced declining sales trends.

By July 2021, only Nougat, Pudding, Geplak, and Marshmallow sold, each with one order.

CUSTOMER & REGIONAL

7. Customers contribution to total sales, and revenue concentrated among top customers

Top 5 Customers (2020):

- Prasco Group: $219K
- Ei: $189K
- Mylan Corp: $173K
- SSS Group: $169K
- Dharma Ltd.: $151K

Top 5 Customers (2021):

- Ohio: $88K
- Wuxi Group: $88K
- Procter Corp: $75K
- Capweld: $72K
- Nipro: $70K

8. Sales Performance By Regions

- 2020: West ($2.3M), South ($2.0M), Midwest ($1.9M), Northeast ($0.9M).
- 2021: West ($1.1M), South ($0.98M), Midwest ($0.91M), Northeast ($0.47M).

INSIGHTS AND BUSINESS IMPLICATIONS

1. Overall Business Health & Demand Trends

2020 was a strong revenue year ($7.08M), characterized by fluctuating but generally resilient demand, with clear recovery phases after downturns (notably August and Q4).

2021 shows a sharp contraction in sales performance (−51% YoY revenue decline), signalling significant business disruption, demand erosion, or incomplete data capture beyond July.

The −99% drop in July 2021 across all KPIs strongly suggests operational shutdown, data truncation, or major supply-chain interruption, rather than organic demand decline.

Implication: The business transitioned from a growth-capable state in 2020 into a high-risk decline phase in 2021, requiring urgent diagnostic investigation into operations, supply, and data completeness.

2. Revenue Drivers & Product Dependence

Sales revenue is highly concentrated in a small group of products, led consistently by Chocolate Truffle and Nougat across both years.

Despite overall decline in 2021, top products remained unchanged, indicating that demand preference did not shift — availability or execution likely did.

By July 2021, product diversity collapsed (only 4 products sold, 1 order each), signaling severe portfolio underutilization.

Implication: Velora Stores is over-reliant on a narrow product set, exposing the business to revenue shocks. Diversification and lifecycle management of lower-performing products are critical.

3. Pricing Strategy & Customer Value (AOV Analysis)

Average Order Value dropped sharply from $928.90 (2020) to $456.67 (2021) (−51%). The entire AOV distribution shifted downward, indicating:

- Smaller basket sizes

- Reduced purchasing power or discounting

- Loss of high-value customers

Implication: The decline is not only volume-driven but also value-driven, suggesting weakened pricing power, customer downgrading, or loss of premium buyers.

4. Sales Channels Performance

All sales channels declined proportionally from 2020 to 2021, with AXW291 consistently outperforming others. No channel showed resilience or counter-cyclical growth.

Implication: The issue is systemic rather than channel-specific. Optimizing channels alone will not reverse decline without addressing upstream demand, supply, or customer engagement.

5. Volume vs Revenue Dynamics

While sales volume and revenue are positively correlated, higher quantities do not guarantee higher revenue.

Example comparisons show pricing differences and product mix significantly affect revenue outcomes.

Implication: Revenue growth should prioritize high-margin orders, not just high-volume sales. 3Pricing optimization and product bundling strategies could improve revenue efficiency.

6. Product Decline & Portfolio Risk

All products declined in 2021, with near-total collapse by July. This suggests operational failure or market exit, not selective product underperformance.

Implication: Immediate intervention is needed to determine whether the decline stems from:

- Supply chain breakdown

- Inventory shortages

- Market withdrawal

- Incomplete or cut-off data

7. Customer Concentration & Revenue Risk

Revenue is heavily concentrated among top customers in both years. Top customers changed entirely from 2020 to 2021, indicating low customer retention at the high-value level.

Implication: The business faces customer concentration risk and weak long-term customer loyalty. Retention strategies for high-value clients are essential.

8. Regional Performance Patterns

West and South regions consistently outperform, while the Northeast underperforms across both years. Despite overall decline, regional rankings remained stable, suggesting structural regional differences.

Implication: Marketing and expansion investments should prioritize high-yield regions, while underperforming regions require targeted diagnostics or cost rationalization.

Recommendations

Based on historical sales trends and performance patterns, the following recommendations are proposed to stabilize performance and drive sustainable growth at Velora Stores:

Velora Stores should first validate operational continuity and data completeness, particularly around the sharp performance decline observed in mid-2021. Confirming whether the decline was driven by operational disruption, inventory shortages, or data limitations is essential before implementing growth initiatives.

To strengthen revenue stability, the business should prioritize its core revenue-generating products while reducing over-reliance on a narrow product portfolio. Product bundling, selective repricing, and portfolio rationalization can help improve revenue consistency and customer purchase depth.

Given the significant decline in Average Order Value, the company should implement pricing and basket-expansion strategies such as bundled offers, minimum order thresholds, and targeted promotions aimed at recovering transaction value without excessive discounting.

Sales growth efforts should shift from pure volume expansion to revenue quality optimization, focusing on high-margin customers, products, and channels. Channel performance analysis indicates that resources should be concentrated on consistently high-performing channels while underperforming channels are optimized or streamlined.

Customer retention should be treated as a strategic priority. The concentration of revenue among a small group of customers and the turnover in top buyers between years highlight the

need for structured retention programs for high-value customers to reduce volatility and improve lifetime value.

Regionally, investments should be aligned with demand strength by scaling operations in high-performing regions and conducting targeted diagnostics in underperforming territories to identify pricing, distribution, or demand barriers.

APPENDIX 1: TABLES

	OrderNumber	OrderDate	CustomerCode	CustomerName	Customer Type	WarehouseCode	DeliveryName	DeliveryRegion	OrderStatus	ProductDescription	OrderQuantity	UnitPrice	LineTotal
36	SO - 000879	29/12/2020	Medline	Medline	Wholesale	GUT930	Medline	NV	Completed	Chocolate Rugelach	45	34.99	1,574.51
37	SO - 0001460	29/12/2020	Qualitest	Qualitest	Club	NXH382	Qualitest	MD	Completed	Chocolate	24	25.05	601.26
38	SO - 0001603	29/12/2020	Sundial	Sundial	Online	FLR025	Sundial	MO	Completed	Geplak	51	26.58	1,355.46
39	SO - 0001975	29/12/2020	Nipro	Nipro	Online	AXW291	Nipro	VA	Completed	Pudding	27	35.1	947.64
40	SO - 0002341	29/12/2020	OHTA'S Corp	OHTA'S Corp	Club	AXW291	OHTA'S Corp	TX	Completed	Chocolate Truffle	23	185.09	4,257.14
41	SO - 0002840	29/12/2020	Rochester Ltd	Rochester Ltd	Wholesale	NXH382	Rochester Ltd	NH	Completed	Rice Krispie treats	28	14.42	403.65
42	SO - 0002870	29/12/2020	WakeFern	WakeFern	Wholesale	AXW291	WakeFern	IL	Completed	S'mores	44	11.57	509.28
43	SO - 0003568	29/12/2020	Amylin Group	Amylin Group	Online	AXW291	Amylin Group	NH	Completed	Rice Krispie treats	32	15.73	503.5
44	SO - 0003999	29/12/2020	OUR Ltd	OUR Ltd	Club	AXW291	OUR Ltd	KY	Completed	Panforte	15	30.3	454.43
45	SO - 0004007	29/12/2020	OUR Ltd	OUR Ltd	Club	AXW291	OUR Ltd	UT	Deleted	S'mores	41	10.21	418.77
46	SO - 0004931	29/12/2020	Amylin Group	Amylin Group	Wholesale	GUT930	Amylin Group	FL	Completed	Fudge	24	33	791.97
47	SO - 0005336	29/12/2020	U.S. Ltd	U.S. Ltd	Wholesale	GUT930	U.S. Ltd	NV	Completed	Panforte	45	33.02	1,485.73
48	SO - 0005454	29/12/2020	Uriel Group	Uriel Group	Wholesale	AXW291	Uriel Group	VA	Completed	Pudding	41	36.74	1,506.30
49	SO - 0005816	29/12/2020	Mylan Corp	Mylan Corp	Wholesale	GUT930	Mylan Corp	IN	Completed	Nougat	36	88.38	3,181.60
50	SO - 0005899	29/12/2020	OUR Ltd	OUR Ltd	Export	AXW291	OUR Ltd	NV	Completed	Rice Krispie treats	51	15.54	792.68
51	SO - 0006226	29/12/2020	OUR Ltd	OUR Ltd	Export	GUT930	OUR Ltd	IN	Completed	Rice Krispie treats	31	16.22	502.84
52	SO - 0006732	29/12/2020	Rochester Ltd	Rochester Ltd	Wholesale	AXW291	Rochester Ltd	ID	Completed	Panforte	28	30.21	845.88
53	SO - 0007019	29/12/2020	Ohio	Ohio	Wholesale	AXW291	Ohio	IN	Deleted	Rice Krispie treats	48	16.11	773.11
54	SO - 0007768	29/12/2020	Mylan Corp	Mylan Corp	Wholesale	FLR025	Mylan Corp	VA	Completed	Brittle	43	66.46	2,857.75
55	SO - 0007897	29/12/2020	Exact-Rx, Corp	Exact-Rx, Corp	Distributor	FLR025	Exact-Rx, Corp	MS	Completed	Marshmallow	34	7.47	253.92
56	SO - 000148	28/12/2020	E. Ltd	E. Ltd	Distributor	NXH382	E. Ltd	MD	Completed	Chocolate	49	26.56	1,301.46
57	SO - 0001021	28/12/2020	Ascend Ltd	Ascend Ltd	Distributor	AXW291	Ascend Ltd	NH	Completed	Chocolate Rugelach	41	34.22	1,402.97
58	SO - 0001544	28/12/2020	New Ltd	New Ltd	Wholesale	FLR025	New Ltd	UT	Completed	S'mores	32	11.76	376.21
59	SO - 0002458	28/12/2020	Apotheca, Ltd	Apotheca, Ltd	Wholesale	NXH382	Apotheca, Ltd	NH	Completed	Nougat	46	84.35	3,880.04
60	SO - 0002560	28/12/2020	Medsep Group	Medsep Group	Distributor	GUT930	Medsep Group	TN	Completed	Turkish delight	48	60.58	2,907.71
61	SO - 0002955	28/12/2020	3LAB, Ltd	3LAB, Ltd	Club	AXW291	3LAB, Ltd	CA	Completed	Brittle	16	66.91	1,070.56
62	SO - 0003224	28/12/2020	OUR Ltd	OUR Ltd	Distributor	AXW291	OUR Ltd	ND	Completed	Turkish delight	18	71.46	1,286.19
63	SO - 0003227	28/12/2020	OUR Ltd	OUR Ltd	Online	AXW291	OUR Ltd	MO	Completed	Marzipan	33	32.91	1,085.88
64	SO - 0003249	28/12/2020	Ohio	Ohio	Wholesale	GUT930	Ohio	KS	Completed	Brittle	19	60.1	1,141.83
65	SO - 0003320	28/12/2020	Dharma Ltd	Dharma Ltd	Wholesale	GUT930	Dharma Ltd	UT	Completed	Chocolate Rugelach	30	38.53	1,155.80
66	SO - 0004664	28/12/2020	Qualitest	Qualitest	Wholesale	AXW291	Qualitest	CA	Deleted	Chocolate Rugelach	24	34.49	827.67
67	SO - 0004816	28/12/2020	E. Ltd	E. Ltd	Wholesale	FLR025	E. Ltd	OR	Completed	Turkish delight	26	63.01	1,638.29
68	SO - 0007683	28/12/2020	Dharma Ltd	Dharma Ltd	Wholesale	GUT930	Dharma Ltd	IL	Completed	Rice Krispie treats	23	14.88	342.27
69	SO - 0007797	28/12/2020	Apotheca, Ltd	Apotheca, Ltd	Wholesale	AXW291	Apotheca, Ltd	KY	Completed	Chocolate Rugelach	49	33.01	1,617.57

Fig 1. Customers Data

Customer_ID	CustomerName	Customer Type	DeliveryName
0	21st Ltd	Distributor	21st Ltd
1	3LAB, Ltd	Wholesale	3LAB, Ltd
2	Amylin Group	Online	Amylin Group
3	Apollo Ltd	Wholesale	Apollo Ltd
4	Apotheca, Ltd	Wholesale	Apotheca, Ltd
5	Ascend Ltd	Distributor	Ascend Ltd
6	AuroMedics Corp	Wholesale	AuroMedics Corp
7	Avon Corp	Wholesale	Avon Corp
8	Bare	Club	Bare
9	Burt's Corp	Wholesale	Burt's Corp
10	Capweld	Wholesale	Capweld
11	Dharma Ltd	Wholesale	Dharma Ltd
12	E. Ltd	Wholesale	E. Ltd
13	ETUDE Ltd	Wholesale	ETUDE Ltd
14	Ei	Wholesale	Ei
15	Elorac, Corp	Wholesale	Elorac, Corp
16	Eminence Corp	Distributor	Eminence Corp
17	Exact-Rx, Corp	Distributor	Exact-Rx, Corp
18	Fenwal, Corp	Online	Fenwal, Corp
19	Linde	Distributor	Linde
20	Llorens Ltd	Club	Llorens Ltd
21	Medline	Online	Medline
22	Medsep Group	Club	Medsep Group
23	Mylan Corp	Wholesale	Mylan Corp
24	New Ltd	Wholesale	New Ltd
25	Niconovum Corp	Online	Niconovum Corp
26	Nipro	Online	Nipro
27	O.E. Ltd	Club	O.E. Ltd

Table: Customer_Dim (50 rows)

Fig 2 Customer Dimension Table

```
1 Date = CALENDARAUTO()
```

Date	Year	Month NO.	Quarter	Month	Month Short
01/01/2020	2020	1	Q1	January	Jan
02/01/2020	2020	1	Q1	January	Jan
03/01/2020	2020	1	Q1	January	Jan
04/01/2020	2020	1	Q1	January	Jan
05/01/2020	2020	1	Q1	January	Jan
06/01/2020	2020	1	Q1	January	Jan
07/01/2020	2020	1	Q1	January	Jan
08/01/2020	2020	1	Q1	January	Jan
09/01/2020	2020	1	Q1	January	Jan
10/01/2020	2020	1	Q1	January	Jan
11/01/2020	2020	1	Q1	January	Jan
12/01/2020	2020	1	Q1	January	Jan
13/01/2020	2020	1	Q1	January	Jan
14/01/2020	2020	1	Q1	January	Jan
15/01/2020	2020	1	Q1	January	Jan
16/01/2020	2020	1	Q1	January	Jan
17/01/2020	2020	1	Q1	January	Jan
18/01/2020	2020	1	Q1	January	Jan
19/01/2020	2020	1	Q1	January	Jan
20/01/2020	2020	1	Q1	January	Jan
21/01/2020	2020	1	Q1	January	Jan
22/01/2020	2020	1	Q1	January	Jan
23/01/2020	2020	1	Q1	January	Jan
24/01/2020	2020	1	Q1	January	Jan
25/01/2020	2020	1	Q1	January	Jan
26/01/2020	2020	1	Q1	January	Jan
27/01/2020	2020	1	Q1	January	Jan
28/01/2020	2020	1	Q1	January	Jan

Table: Date (731 rows)

Fig 3: Date Table

OrderNumber	WarehouseCode	OrderStatus	ProductDescription
SO - 0003201	AXW291	Completed	Geplak
SO - 0004015	AXW291	Completed	Geplak
SO - 0007370	AXW291	Completed	Geplak
SO - 0001736	AXW291	Completed	Geplak
SO - 0003057	AXW291	Completed	Geplak
SO - 0004632	AXW291	Completed	Geplak
SO - 0001446	AXW291	Completed	Geplak
SO - 0001289	AXW291	Completed	Geplak
SO - 0007234	AXW291	Completed	Geplak
SO - 000874	AXW291	Completed	Geplak
SO - 0002358	AXW291	Completed	Geplak
SO - 0004916	AXW291	Completed	Geplak
SO - 0006006	AXW291	Completed	Geplak
SO - 000242	AXW291	Completed	Geplak
SO - 000263	AXW291	Completed	Geplak
SO - 000827	AXW291	Completed	Geplak
SO - 0006282	AXW291	Completed	Geplak
SO - 0006813	AXW291	Completed	Geplak
SO - 0004819	AXW291	Completed	Geplak
SO - 0003802	AXW291	Completed	Geplak
SO - 0004631	AXW291	Completed	Geplak
SO - 0001157	AXW291	Completed	Geplak
SO - 0008099	AXW291	Completed	Geplak
SO - 0004585	AXW291	Completed	Geplak
SO - 0004161	AXW291	Completed	Geplak
SO - 0005083	AXW291	Completed	Geplak
SO - 0003561	AXW291	Completed	Geplak
SO - 0006642	AXW291	Completed	Geplak

Table: Order_Dim (8,000 rows)

Fig 4: Order Dimension Table

Customer_ID	OrderNumber	OrderDate	DeliveryRegion	OrderQuantity	UnitPrice	LineTotal
1	SO - 0001037	Wednesday, 30 December 2020	UT	45	$35.14	$1,581.18
8	SO - 0001363	Wednesday, 30 December 2020	UT	48	$35.93	$1,724.75
29	SO - 0004007	Tuesday, 29 December 2020	UT	41	$10.21	$418.77
24	SO - 0001544	Monday, 28 December 2020	UT	32	$11.76	$376.21
11	SO - 0003320	Monday, 28 December 2020	UT	30	$38.53	$1,155.80
15	SO - 0005943	Saturday, 26 December 2020	UT	45	$82.31	$3,703.95
46	SO - 0001750	Thursday, 24 December 2020	UT	24	$27.46	$658.94
1	SO - 0003119	Thursday, 24 December 2020	UT	32	$10.24	$327.55
49	SO - 0004277	Wednesday, 23 December 2020	UT	32	$38.77	$1,240.70
18	SO - 0006600	Wednesday, 23 December 2020	UT	26	$93.84	$2,439.73
18	SO - 0001886	Tuesday, 22 December 2020	UT	37	$61.45	$2,273.52
5	SO - 000488	Sunday, 20 December 2020	UT	31	$30.89	$957.62
37	SO - 0004572	Saturday, 19 December 2020	UT	47	$67.09	$3,153.36
16	SO - 0004935	Saturday, 19 December 2020	UT	40	$39.48	$1,579.31
27	SO - 0001450	Friday, 18 December 2020	UT	26	$40.11	$1,042.86
41	SO - 0007024	Thursday, 17 December 2020	UT	17	$16.86	$286.62
7	SO - 0003105	Wednesday, 16 December 2020	UT	24	$25.23	$605.47
17	SO - 0001519	Tuesday, 15 December 2020	UT	27	$30.90	$834.27
4	SO - 0005446	Tuesday, 15 December 2020	UT	33	$69.90	$2,306.57
27	SO - 0006524	Tuesday, 15 December 2020	UT	39	$19.23	$749.91
48	SO - 0004093	Monday, 14 December 2020	UT	25	$63.64	$1,591.10
46	SO - 0005341	Saturday, 12 December 2020	UT	15	$6.13	$91.89
48	SO - 000450	Thursday, 10 December 2020	UT	16	$60.49	$967.89
28	SO - 0003410	Thursday, 10 December 2020	UT	43	$63.07	$2,712.14
20	SO - 0005140	Thursday, 10 December 2020	UT	34	$40.53	$1,378.05
25	SO - 0004793	Sunday, 6 December 2020	UT	51	$27.29	$1,391.67
11	SO - 0001565	Saturday, 5 December 2020	UT	19	$90.98	$1,728.63
22	SO - 0005689	Saturday, 5 December 2020	UT	15	$25.38	$380.76

Table: Order_Fact (8,000 rows)

Fig 5: Order Fact Table

Name	Status	Short Code	Division	Region
Alabama	State	AL	East South Central	South
Alaska	State	AK	Pacific	West
Arizona	State	AZ	Mountain	West
Arkansas	State	AR	West South Central	South
California	State	CA	Pacific	West
Colorado	State	CO	Mountain	West
Connecticut	State	CT	New England	Northeast
Delaware	State	DE	South Atlantic	South
District of Columbia	Federal district	DC	South Atlantic	South
Florida	State	FL	South Atlantic	South
Georgia	State	GA	South Atlantic	South
Hawaii	State	HI	Pacific	West
Idaho	State	ID	Mountain	West
Illinois	State	IL	East North Central	Midwest
Indiana	State	IN	East North Central	Midwest
Iowa	State	IA	West North Central	Midwest
Kansas	State	KS	West North Central	Midwest
Kentucky	State (Commonwealth)	KY	East South Central	South
Louisiana	State	LA	West South Central	South
Maine	State	ME	New England	Northeast
Maryland	State	MD	South Atlantic	South
Massachusetts	State (Commonwealth)	MA	New England	Northeast
Michigan	State	MI	East North Central	Midwest
Minnesota	State	MN	West North Central	Midwest
Mississippi	State	MS	East South Central	South
Missouri	State	MO	West North Central	Midwest
Montana	State	MT	Mountain	West
Nebraska	State	NE	West North Central	Midwest

Table: Region (51 rows)

Fig 6: Region Table

APPENDIX 2: MODELS

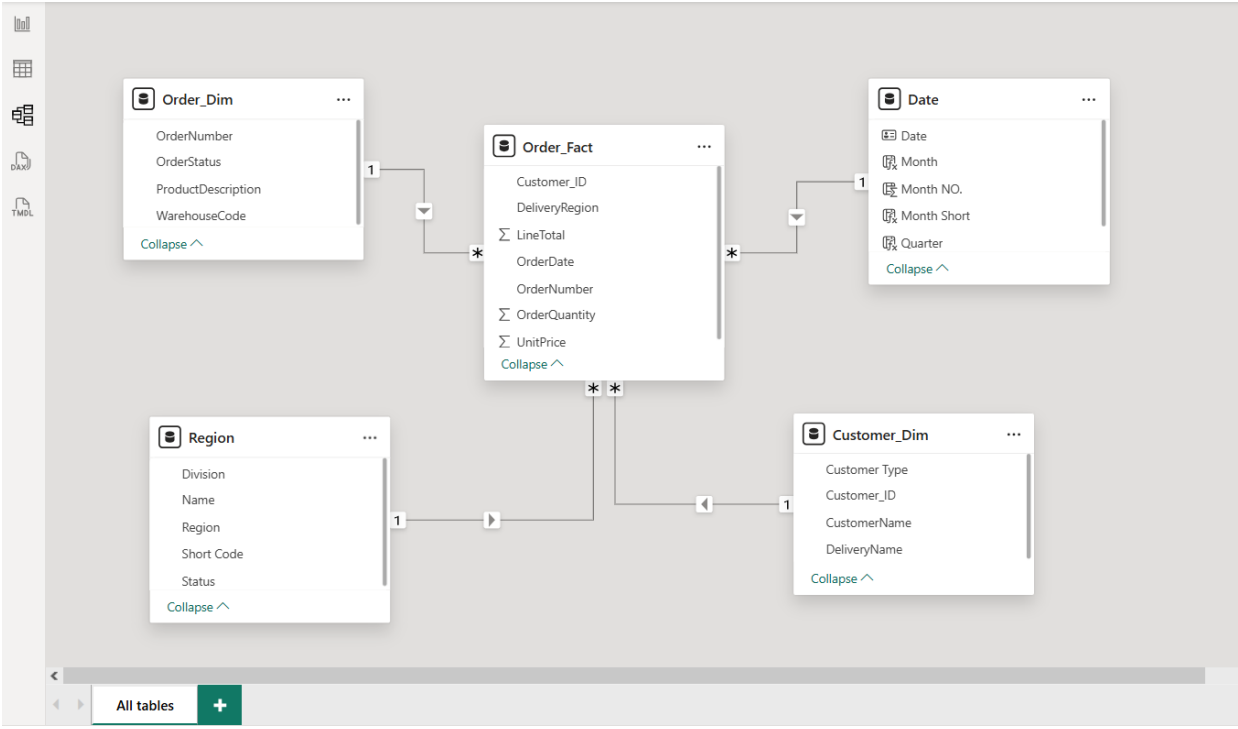

Fig 7: Data Modelling (Star Schema)

APPENDIX 3: DAX MEASURES

1. AOV

- AOV_PM = CALCULATE([Average Order Value (AOV)], DATEADD('Date'[Date],-1,MONTH))
- AOV_VAR = [Average Order Value (AOV)]-[AOV_PM]
- Average Order Value (AOV) = CALCULATE(DIVIDE([Total Revenue],
  DISTINCTCOUNT(Order_DIM[OrderNumber])),
  Order_DIM[OrderStatus]="Completed")
- MoM_AOV =

```
VAR Positive_Arrow = UNICHAR(9650) // UP ARROW
VAR Negative_Arrow = UNICHAR(9660)  // Down ARROW
VAR AOV_ = [Average Order Value (AOV)]
VAR PMAOV = [AOV_PM]
VAR Growth = DIVIDE(AOV_-PMAOV,PMAOV,0)
RETURN
 IF (Growth > 0, Positive_Arrow, Negative_Arrow)& "  " & ROUND(Growth*100,2) & "%"
```

- MoM_AOV Color = IF([MoM_AOV Growth]>0, "Green",
  IF([MoM_AOV Growth]<0, "Red",
  "#D0CBCB"))
- MoM_AOV Growth = DIVIDE([Average Order Value (AOV)]-[AOV_PM],[AOV_PM],0)
- Customer Rank = RANKX(ALLSELECTED(Customer_Dim[CustomerName]), [Total Revenue],,
  DESC,Dense)

2. ORDER

- Cmpltd Order = CALCULATE(COUNT(Order_Fact[OrderNumber]),
  Order_Dim[OrderStatus]="Completed")
- Cmpltd Order Rate = DIVIDE([Cmpltd Order],[Total Order],0)
- Deleted Order Rate = 1-[Cmpltd Order Rate]
- MoM_Order =
  ```
  VAR Positive_Arrow = UNICHAR(9650) // UP ARROW
  VAR Negative_Arrow = UNICHAR(9660)  // Down ARROW
  VAR Order_ = [Total Order]
  VAR PMOrder = [ORDER_PM]
  VAR Growth = DIVIDE(Order_-PMOrder,PMOrder,0)
  RETURN
          IF (Growth > 0, Positive_Arrow, Negative_Arrow)& "  " & ROUND(Growth*100,2)
  & "%"
  ```
- MoM_Order Color = IF([MoM_Order Growth]>0, "Green",
  IF([MoM_Order Growth]<0, "Red",
  "#D0CBCB"))
- MoM_Order Growth = DIVIDE([Total Order]-[ORDER_PM],[ORDER_PM],0)
- ORDER_PM = CALCULATE([Total Order],DATEADD('Date'[Date],-1,MONTH))
- ORDER_VAR = [Total Order]-[ORDER_PM]
- Total Order = DISTINCTCOUNT(Order_Fact[OrderNumber])

3. QUANTITY

- MoM_Qty =
  ```
  VAR Positive_Arrow = UNICHAR(9650) // UP ARROW
  VAR Negative_Arrow = UNICHAR(9660)  // Down ARROW
  VAR Qty_ = [Total Qauntity]
  VAR PMQty = [PM_QTY]
  VAR Growth = DIVIDE(Qty_-PMQty,PMQty,0)
  RETURN
          IF (Growth > 0, Positive_Arrow, Negative_Arrow)& "  " & ROUND(Growth*100,2) &
  "%"
  ```
- MoM_Qty Color = IF([MoM_Qty Growth]>0, "Green",
  IF([MoM_Qty Growth]<0, "Red",
  "#D0CBCB"))
- MoM_Qty Growth = DIVIDE([Total Qauntity]-[PM_QTY],[PM_QTY],0)
- PM_QTY = CALCULATE([Total Qauntity],DATEADD('Date'[Date],-1,MONTH))
- QTY_VAR = [Total Qauntity]-[PM_QTY]
- Total Qauntity =
  CALCULATE(SUM(Order_Fact[OrderQuantity]),Order_Dim[OrderStatus]="Completed")

4. REVENUE

- MoM_Rev =
  ```
  VAR Positive_Arrow = UNICHAR(9650) // UP ARROW
  VAR Negative_Arrow = UNICHAR(9660)  // Down ARROW
  VAR Rev_ = [Total Revenue]
  VAR PMRev = [PM_Rev]
  VAR Growth = DIVIDE(Rev_-PMRev,PMRev,0)
  ```

```
RETURN
        IF (Growth > 0, Positive_Arrow, Negative_Arrow)& "  " & ROUND(Growth*100,2) &
"%"
```

- ```
  MoM_Rev Color = IF([MoM_Rev Growth]>0, "Green",
      IF([MoM_Rev Growth]<0, "Red",
      "#D0CBCB"))
  ```
- ```
  MoM_Rev Growth = DIVIDE([Total Revenue]-[PM_REV],[PM_REV],0)
  ```
- ```
  PM_REV = CALCULATE([Total Revenue],DATEADD('Date'[Date],-1,MONTH))
  ```
- ```
  REV_VAR = [Total Revenue]-[PM_REV]
  ```
- ```
  Total Revenue =
  CALCULATE(SUM(Order_Fact[LineTotal]),Order_Dim[OrderStatus]="Completed")
  ```
  ```